

Solar Cannibalisation Spread

STRUCTURAL

Sell the midday block, buy the evening block, in the day-ahead auction

Jan 2019 – Jun 2026

Source: EPEX SDAC day-ahead auctions via Energy-Charts (Fraunhofer ISE, CC BY 4.0). Backtest by author.

2025 MAR–OCT P&L

+104

€/MWh, DE-LU

WIN RATE MAR–OCT '25

100%

DE-LU

7-YR MAR–OCT AVG

+70

€/MWh, DE-LU

2025 MAR–OCT P&L

+107

€/MWh, CZ

WORST DAY (2025)

-84

€/MWh, DE-LU

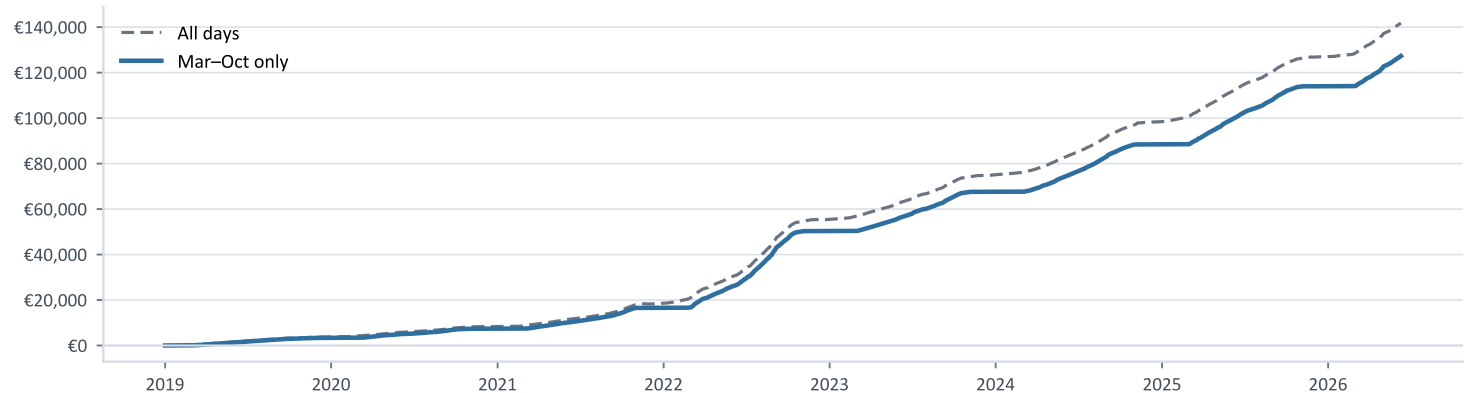
STRATEGY & EDGE

The same physics a battery monetises, expressed without owning one. Each day in the day-ahead auction (or via shaped block futures) the strategy sells the 11:00–15:00 block and buys the 18:00–21:00 block in equal MW — a pure intra-day shape position, flat on net volume.

The P&L is the evening-minus-midday price difference. It is structural because solar must run midday and bids at any price to clear, while the evening ramp is met by gas and inflexible demand. As installed solar grows the midday block is increasingly cannibalised, so the spread trends wider — and the March–October solar season concentrates almost all of it.

Cumulative P&L per MWh of block size (DE-LU)

All days vs the March–October solar season only



HOW IT WORKS

- Short leg: average of delivery hours 11–14; long leg: average of hours 18–20. Equal size, settled at cleared prices.
- P&L = evening block – midday block, minus 0.30 €/MWh per leg in fees and slippage.
- Volume-neutral: no directional power price exposure, only the intra-day shape.
- Mar–Oct win rate has been 100% for four consecutive years — a persistent structural imbalance, not a fluke.
- Persists because expressing it at size needs exchange/BRP access (see access ladder in the research note).
- Tail risk is a dark, wind-starved winter evening with a midday price spike — rare but –80 to –90 €/MWh days exist.

ASSUMPTIONS & SPECIFICATION

Equal-MW short 11–15h / long 18–21h, settled daily at day-ahead clearing prices · transaction cost 0.30€/MWh per leg (0.60 €/MWh round trip) · P&L quoted per MWh of block size so it scales with position · no overnight inventory, no balancing exposure · 'Mar–Oct' isolates the solar season and is a calendar filter known in advance — no look-ahead.

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Performance detail, seasonality and risk

Jan 2019 – Jun 2026

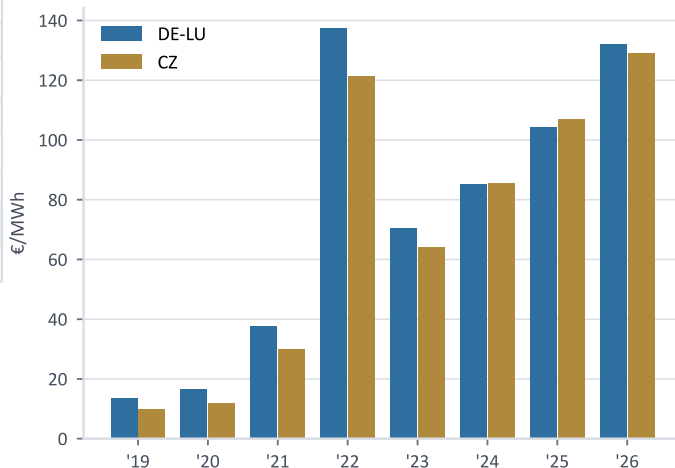
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PERFORMANCE BY YEAR (MARCH–OCTOBER)

Year	DE P&L	Win	Sharpe	CZ P&L
2019	+13.6	96%	22.8	+9.8
2020	+16.5	96%	20.1	+12.0
2021	+37.8	100%	25.0	+30.0
2022	+137.4	100%	30.2	+121.5
2023	+70.5	100%	34.2	+64.0
2024	+85.1	100%	39.2	+85.7
2025	+104.3	100%	39.7	+107.1
2026 YTD	+132.0	100%	40.8	+129.0

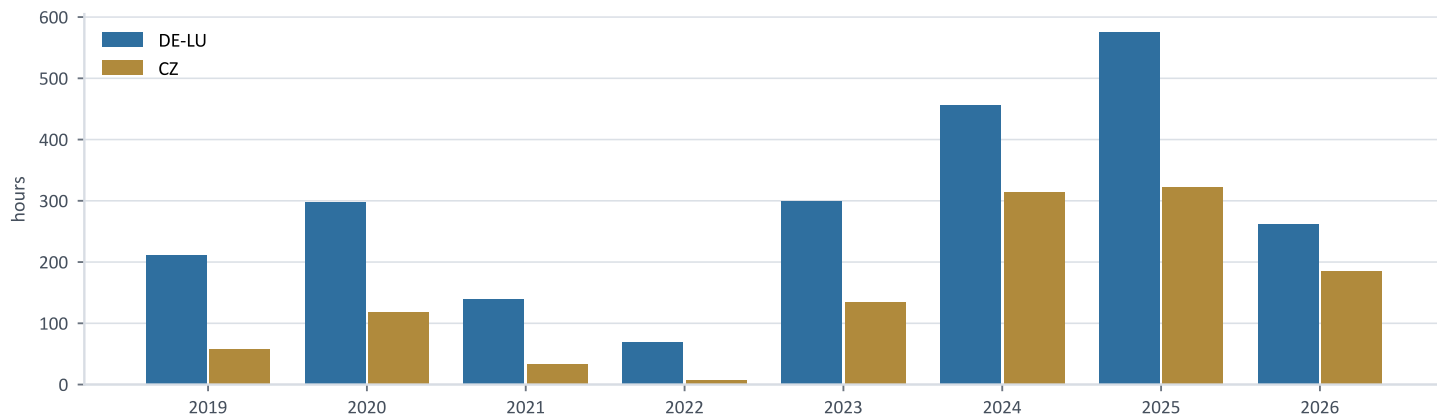
P&L in €/MWh of block. Sharpe is gross and per-MWh, annualised from daily spreads — it overstates realisable risk-adjusted return (ignores capacity limits, access cost and treats days as independent). Read avg P&L and win rate as the primary metrics.

Avg seasonal spread by year



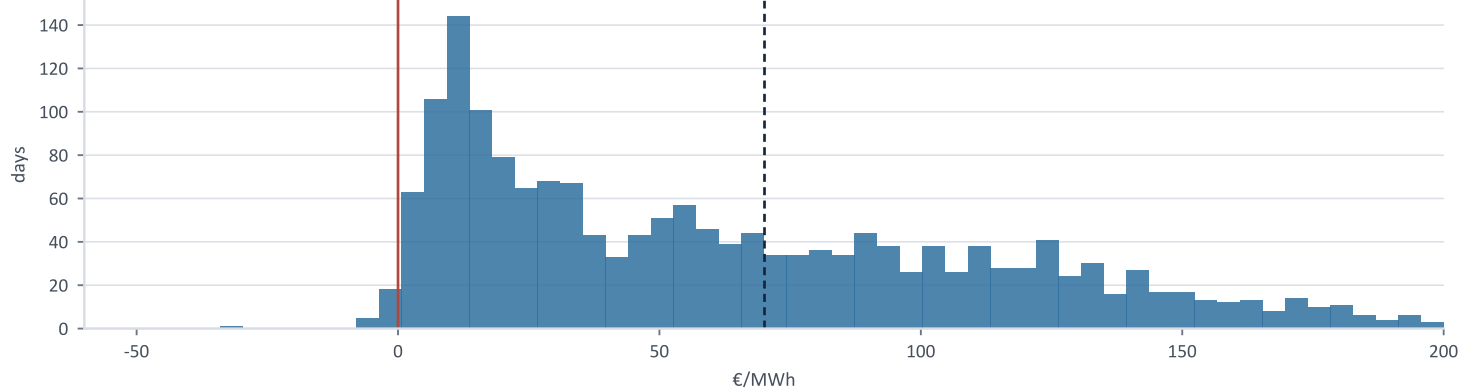
Why the edge grows: negative-price hours per year

Hours with negative day-ahead prices — the symptom of solar over-supply



Daily P&L distribution, Mar–Oct (DE-LU)

EUR/MWh, axis -60 to 200 (67 richer days off-axis); the structural skew is to the right



KEY RISKS & EDGE DECAY

Same decay clock as the battery trade: large-scale storage and demand-shifting eventually arbitrage the midday-evening spread away— the annual bars are the early-warning gauge. · Access is the binding constraint, not the signal: capturing it at size requires exchange membership or a BRP umbrella, whose costs must stay below the spread. · Winter tail: a low-solar evening peak with a midday spike inverts the trade (worst days ≈ -85 €/MWh). · Off-season (Nov–Feb) the edge is thin — the calendar filter is doing real work.